Data Manipulation and Matrix Operations in R

Computing Fundamentals for Social Sciences

Claudia Pigini

Outline

Scalars and Vectors

- **Scalar**: a single value, e.g. number or string.
- **Vector**: ordered collection of elements of the same type.

```
# Scalar
a <- 5

# Vector
v <- c(1, 3, 5, 7)
```

Matrices

- 2-dimensional rectangular data structure.
- All elements must be of the same type.
- Defined by number of rows and columns.

```r
# Create a 2x3 matrix
m <- matrix(1:6, nrow = 2, ncol = 3)
```

$$m = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$$

Arrays

- Generalization of matrices to higher dimensions (3D, 4D, ...).
- All elements must be same type.

```
# Create a 3D array: 2x2x2
arr <- array(1:8, dim = c(2,2,2))
```

Vector Arithmetic

```
x <- c(1, 2, 3)
y <- c(4, 5, 6)

# Element-wise addition
x + y   # returns c(5, 7, 9)

# Element-wise multiplication
x * y   # returns c(4, 10, 18)

# Scalar multiplication
2 * x   # returns c(2, 4, 6)
```

Matrix Transposition

- Flip rows and columns.
- Denoted A^T.

```r
m <- matrix(1:6, nrow = 2)
t(m)
```

$$m = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}, \quad m^T = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

Row and Column Operations

- Extract rows and columns with indexing.
- Modify specific rows or columns.

```
m <- matrix(1:6, nrow=2)

# Extract first row
m[1, ]

# Extract second column
m[, 2]

# Set second row to zeros
m[2, ] <- 0
```

Row and Column Sums

- Compute sums across rows or columns easily.

```
m <- matrix(1:6, nrow=2)

# Sum of each row
rowSums(m)   # returns c(9, 12)

# Sum of each column
colSums(m)   # returns c(3, 7, 11)
```

Matrix Addition and Scalar Multiplication

```r
A <- matrix(1:4, 2, 2)
B <- matrix(c(5, 6, 7, 8), 2, 2)

# Matrix addition
C <- A + B

# Scalar multiplication
D <- 3 * A
```

Matrix Multiplication

- Different from element-wise multiplication.
- Follows linear algebra rules: $(AB)_{ij} = \sum_k A_{ik} B_{kj}$.

```
A <- matrix(1:6, 2, 3)
B <- matrix(7:12, 3, 2)

# Matrix product
C <- A %*% B
```

Element-wise Multiplication

```
A <- matrix(1:4, 2, 2)
B <- matrix(c(5, 6, 7, 8), 2, 2)

# Element-wise product
C <- A * B
```

Summary

- Scalars, vectors, matrices, and arrays are core R data structures.
- Vector operations are element-wise.
- Matrix operations include transpose, addition, scalar multiplication, and matrix multiplication.
- Use `rowSums()` and `colSums()` to quickly get sums over rows or columns.
- Indexing allows manipulation of rows and columns.

Exercises

1. Create a vector of length 5 and multiply it by 3.
2. Create a 3x3 matrix and extract its second column.
3. Transpose a 2x4 matrix.
4. Calculate row sums and column sums of a matrix.
5. Perform matrix multiplication of two conformable matrices.
6. Modify the first row of a matrix to be all ones.

Solutions

```r
# 1.
v <- c(1,2,3,4,5)
v * 3

# 2.
m <- matrix(1:9, 3, 3)
m[,2]

# 3.
m2 <- matrix(1:8, 2, 4)
t(m2)

# 4.
m <- matrix(1:6, 2, 3)
rowSums(m)
colSums(m)

# 5.
A <- matrix(1:6, 2, 3)
B <- matrix(7:12, 3, 2)
A %*% B

# 6.
m[1, ] <- 1
```